Peizhi Xia

peizhix@uci.edu | +1 (949) 992-7941 | Irvine, CA | LinkedIn: peizhi-xia-72811931a

F-1 | Post-Completion OPT pending (EAD estimated ~08/20/2026) | STEM OPT eligible | Embedded Hardware • PCB Design • Firmware • Power & Sensors

SUMMARY

Embedded electrical engineer with hands-on experience spanning Altium PCB design, DC-DC power conversion, MCU firmware, board bring-up, and sensor-data analysis. Completed PCB workflows from schematic/layout through fabrication and bench debug, and comfortable moving between datasheets, C/C++ test firmware, oscilloscopes, and manufacturing-oriented documentation.

EDUCATION

University of California, Irvine (UCI) - M.S. Electrical Engineering (GPA: 3.42) 09/2024-06/2026

Relevant coursework: Industrial & Power Electronics, Embedded System Modeling (SystemC), Analog IC Design, Micro-Sensors & Actuators, Automation Systems.

Shanghai Jiao Tong University (SJTU) - B.E. Measurement Technology & Instrumentation (GPA: 3.16) 09/2020-06/2024

Relevant coursework: Power Electronics, Embedded Systems, Measurement & Control Circuits, Sensor Theory, Automatic Control.

TECHNICAL SKILLS

Hardware/PCB: Altium Designer, schematic capture/layout, two-sided PCB routing, Gerber/JLCPCB fabrication workflow, DC-DC power and analog sensing, datasheet-driven component selection, TINA-TI, Cadence, soldering/rework.

Embedded/Firmware: C/C++, STM32/Keil, 8051-class MCU, GPIO, ADC/DAC, interrupts/timers, UART, SPI, I2C, MQTT/TCP-IP, state-machine control, board bring-up firmware, fault isolation.

Test/Software: Oscilloscope, DMM, function generator, programmable PSU/e-load; continuity/polarity checks, staged power-up, line/load regulation, ripple and efficiency measurement; Python, MATLAB/Simulink, SystemC, Git, Linux.

SELECTED HARDWARE & EMBEDDED PROJECTS

TPS40200 DC-DC Switching Supply & MCU Monitoring System - SJTU 09/2022-01/2023

- Designed a 10-20 V input, 5 V/1 A regulated supply targeting <=100 mVpp ripple and >=80% full-load efficiency; sized magnetics/capacitance, developed voltage/current feedback, and evaluated operating points in TINA-TI.
- Owned ~80% of the team PCB placement/routing in Altium across voltage-regulation, constant-current, current-sharing, and Cortex-M4 monitoring stages; kept high-di/dt loops compact, separated power/signal returns, protected sense routes, and added test points for staged debug.
- Built and bench-tested the first regulator stage; measured output voltage, line/load regulation, ripple, and efficiency with an oscilloscope, DMM, programmable supply, and load. Documented a subsystem-by-subsystem bring-up plan for the final board.
- The complete multi-stage PCB remained design-only because of course constraints; clearly distinguish this design artifact from the physically tested first-stage prototype.

Photovoltaic Energy Storage & Embedded Control System - SJTU 09/2021-03/2022

- Led system architecture and designed an Altium PCB integrating solar input, TP4056-based single-cell Li-ion charging, 5 V boost power, MCU control, indicators, and relay-switched loads; generated fabrication files, ordered the board from JLCPCB, and assembled the prototype.
- Implemented C/C++ light-based energy-management logic for charge/discharge and load switching; used modular charge/boost subassemblies to accelerate prototype iteration and replacement.
- Performed continuity/polarity checks and staged power-up before connecting the battery and load; isolated a module pinout mismatch and stabilized the power path and control behavior through rail and signal measurements.

Power Electronics Converter Design & Hardware Bring-Up (EECS 267A/267B) - UCI 01/2025-06/2025

- Modeled a 1 kW boost PFC, 5 kW DC-AC inverter, and 15 kW three-phase inverter in MATLAB/Simulink; quantified PF, THD, ripple, and load-dependent behavior (1 kW PFC model: PF 99.97%, THD 2.44%).
- Designed control/filter elements and reusable parameter-sweep and metric-extraction workflows to compare harmonics, ripple, and transient behavior across operating points.
- Brought up and characterized a course-provided SG3524/IR4427 boost-converter PCB: derived duty-cycle targets, hand-wound a ~171 uH inductor, and executed open-/closed-loop tests with oscilloscope, DMM, and bench supply; separately reconstructed the schematic/layout workflow as a self-directed PCB-learning exercise.

Integrated Energy System Control Based on ML & Model Predictive Control - SJTU 03/2023-06/2024

- Developed MATLAB/Simulink studies for an integrated HVAC/energy system, comparing optimization-based and learning-based control approaches under comfort, energy-cost, and operating constraints.
- Converted high-level control decisions into simplified STM32 logic in Keil/C; used UART/GPIO for bring-up and MQTT telemetry for remote monitoring, threshold alerts, and latency/stability checks.

Environmental Monitoring via Hyperspectral Analysis (ICVL) - UCI 06/2025-Present

- Built an end-to-end Python workflow for VNIR/SWIR ingestion, band alignment, interpolation/normalization, radiance-reflectance fitting, MSE-based reliability metrics, and automated Matplotlib reporting; cross-validated HySpex measurements with MODTRAN simulations.
- Modeled B/G/R/IR camera-channel responses as wavelength integrals to derive and assess a detection matrix; improved annotation consistency by ~15% and reduced error by ~8% through thresholding and labeling guidelines (internal evaluation).

ADDITIONAL EXPERIENCE

Course Grader - Antenna Design for Wireless Communication Links, UCI 09/2025-12/2025

- Supported ~50 students on link budgets, Friis transmission, antenna arrays, and beamforming; reviewed technical work and explained calculation/debugging approaches, strengthening RF fundamentals and engineering communication.